

ABSTRACT

Crowd-Sourced Pothole and Road Hazard Detection using Smartphone Sensor Fusion

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ABSTRACT

Road surface defects such as potholes, speed breakers, cracks, and uneven roads are major causes of road accidents, vehicle damage, and traffic congestion. Conventional road inspection methods are manual, time-consuming, costly, and cannot provide continuous monitoring. To address these challenges, this project proposes a Crowd-Sourced Pothole and Road Hazard Detection using Smartphone Sensor Fusion system that automatically detects road hazards using the sensors available in smartphones.

An Android smartphone application is developed to continuously collect data from the device's built-in sensors, including the GPS, accelerometer, and gyroscope. The application records parameters such as geographical location, linear acceleration, and angular motion while the user travels on the road. These sensor readings are synchronized and processed using sensor fusion techniques to accurately identify potholes, road cracks, speed breakers, and other road anomalies while minimizing false detections caused by normal vehicle movements, braking, or turning.

Whenever a road hazard is detected, the application records the GPS coordinates, timestamp, and sensor measurements, and securely uploads the information to a cloud server. Since the system follows a crowd-sourced approach, reports from multiple users traveling on the same road are combined to verify the existence and severity of the detected hazard. This improves the reliability and accuracy of the detection process and reduces false alarms.

The verified road hazard information is displayed on a real-time digital map, allowing road authorities to identify damaged road sections and prioritize maintenance activities. Drivers can also benefit from safer route planning by receiving information about hazardous road conditions in advance. The proposed system is a low-cost, scalable, and efficient solution that utilizes widely available smartphones without requiring additional hardware, making it suitable for smart city applications and intelligent transportation systems.

References

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